

OSDP 2.3 Proposed PIV Enhancements

Security Industry Association

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Changelog

2025-12-10 Consolidated application-level errors on `osdp_PDERROR` (two-byte codes, optional SW1|SW2 for APDU failures, multi-error support); moved the PIV error set to 0x0100–0x01FF and updated PIV commands to use `osdp_PDERROR` (transport faults remain `osdp_NAK`); made the Global Interface Mask in `osdp_PIVMODE` a required byte (no TLV tag) with an updated field layout; clarified that `osdp_CARDSTATUSR` is solicited-only after PD preparation with legacy poll-based `osdp_RAW` as the aggregated card-present indication; and noted the deprecation of `osdp_GENAUTH` with a recommendation to migrate to `osdp_CRAUTH`.

2025-12-04 Clarified that `osdp_CRAUTHR` returns the full GENERAL AUTHENTICATE data including the Dynamic Authentication Template (excluding only the ISO/IEC 7816 status word).

2025-11-12 Added PIV Biometric Match request/reply and tables (one-bit type values only); moved Enhanced PIV NAKs to a 0x20+ base (added 0x27 Credential Not Present/Removed, 0x28 Biometric Type Not Supported; clarified 0x23 Credential Security Status Not Satisfied); aligned CRAUTHR error handling to reference the enhanced NAK set; relocated Enhanced PIV (0x04) and PIV Auto (0x08) indicators to Function Code 12 and cleaned Function Code 13 to interface-only bits (removed non-interface flags; removed PIV Auto from Function Code 14); updated CARDSTATUS gating to Function Code 12 bit 0x04; added a PIV Mode Flags table and clarified that any legacy AID lists cited were from prior proposals (not OSDP 2.2); enabled PDF TOC hyperlinks and improved table/caption pagination.

2025-10-30 Clarified CRAUTH command/reply behaviour (including RSA, AES, and ECC support), overhauled PIV Mode configuration with TLV-based application and interface controls (including PIV Auto), added card status request/reply messages, expanded PIV data retrieval semantics, introduced PIV Put Data, PIV Secure PIN Entry, TWIC privacy-key load, VCI pairing transmit, and VCI trust-anchor load/status commands, and documented the enhanced error-code set (including VCI/PIV Auto conditions).

2025-09-17 Major PIV Mode and SPE revisions.

2025-06-25 Added multi-AID usage and VCI support notes.

2025-05-27 New function code definitions for 2.3 biometric handling.

1 Introduction

This document captures proposed enhancements to the SIA Open Supervised Device Protocol (OSDP) needed to carry FICAM-compliant Personal Identity Verification (PIV) workflows. The material consolidates draft content, reviewer comments, and interoperability feedback collected during the 2025 working sessions and re-expresses it as normative guidance for command authors and device implementers.

The chapters that follow highlight changes required for existing OSDP 2.2 commands, introduce new transactions where no equivalent existed, and provide supplemental examples that reflect current federal credentialing practices. Wherever the proposal diverges from the baseline specification, the text records both the behavioral requirements and the error handling expectations so that access control units (ACUs) and peripheral devices (PDs) can adopt the profile without relying on vendor extensions.

2 Changes to OSDP 2.2

This section documents the normative deltas that apply when an OSDP 2.2 implementation adopts the Enhanced PIV profile. Each subsection references the original command definition and records the precise behavioral updates, encoding clarifications, and new error handling rules introduced by this proposal.

2.1 Updated Command Definitions

2.1.1 Authentication Challenge (osdp_CRAUTH)

OSDP 2.2 transports the ISO/IEC 7816-4 GENERAL AUTHENTICATE command through osdp_CRAUTH. The OSDP 2.3 profile maintains the same framing fields (TOTAL, OFFSET, DATA_LEN) but extends the semantics of the command payload so PDs can carry both the classic RSA signing flow and the enhanced elliptic-curve key-agreement flow.

- The Algorithm field continues to be sent in the first fragment only. Codes 0x05, 0x06, and 0x07 select RSA signing operations and preserve backwards compatibility with OSDP 2.2. Codes 0x08, 0x0A, and 0x0C select AES encryption using 128-, 192-, and 256-bit keys respectively. Codes 0x11 and 0x14 select elliptic-curve key agreement using the P-256 and P-384 curves.
- The Key field retains its role as the card's reference identifier. PDs **SHALL NOT** reinterpret or remap the identifier.
- For RSA algorithms (0x05, 0x06, 0x07), the Challenge field carries the datum to be signed. The PD forwards the value unmodified; padding (for example PKCS#1 v1.5 or PSS) is the responsibility of the ACU.
- For AES algorithms (0x08, 0x0A, 0x0C), the Challenge field contains the plaintext block(s) to be encrypted. PDs **SHALL** send the data to the credential without padding; ACUs **SHALL** provide data aligned to the block size dictated by the selected key length and card profile.
- For algorithms 0x11 and 0x14, the Challenge field conveys the peer public key. The PD **SHALL** issue GENERAL AUTHENTICATE with CLA = 0x00 (or 0x0C when a secure channel is active), INS = 0x87, P1 = Algorithm, and P2 = Reference Identifier. The command data **SHALL** be encoded as 7C Len 82 00 85 Len (04 || X || Y), where X and Y are the uncompressed coordinates for the selected curve and are left-padded with 0x00 as required. The credential performs elliptic-curve Diffie-Hellman in accordance with SP 800-73; the derived secret is returned in the reply fragments described below.

Table 1: Authentication Challenge (osdp_CRAUTH)

Size (bytes)	Name	Meaning	Value
1	CMND	Command identifier.	0xA5
2	TOTAL	Length of the complete message, least-significant byte first.	0x0000–0xFFFF
2	OFFSET	Offset of this fragment within the complete message, least-significant byte first.	0x0000–0xFFFF
2	DATA_LEN	Length of the fragment payload, least-significant byte first.	0x0000–0xFFFF
1	Algorithm	Cryptographic mechanism. Supported values include 0x05/0x06/0x07 (RSA signature), 0x08/0x0A/0x0C (AES encryption with 128/192/256-bit keys), and 0x11/0x14 (ECC-P256/P384 ECDH).	0x00–0xFF
1	Reference Identifier	Key reference passed to GENERAL AUTHENTICATE.	0x00–0xFF
0–n	Challenge	For RSA algorithms, the datum to be signed (no padding applied by the PD). For AES algorithms, the plaintext block(s) to be encrypted, pre-padded by the ACU as required. For 0x11/0x14, the peer public key encoded as 7C Len 82 00 85 Len (04 X Y) with coordinates padded to curve length.	0x00–0xFF

Deprecated osdp_GENAUTH The legacy osdp_GENAUTH command is deprecated and expected to be removed in a future revision of this specification. Using a PIV credential to decrypt or process arbitrary data is unsafe and unnecessary for credential verification; implementations **SHOULD** use osdp_CRAUTH flows instead.

2.1.2 Response to Challenge (osdp_CRAUTHR)

The reply continues to use the OSDP 2.2 multi-part format. Successful execution yields a response containing the credential's entire GENERAL AUTHENTICATE data (minus the ISO/IEC 7816 status word), including the returned Dynamic Authentication Template; PDs **SHALL NOT** strip outer BER-TLV tags or lengths. If the credential reports an error, the PD **SHALL** return osdp_PDERROR with the applicable PIV code—for example 0x0120 (function not supported by credential), 0x0121 (insufficient PD resources), 0x0122 (insufficient credential storage), 0x0123 (credential security status not satisfied), 0x0124 (data object/tag not found), 0x0125 (VCI not established), or 0x0127 (credential not present/removed). Generic parameter/security preconditions may continue to use 0x0005 (incorrect parameter) and 0x0006/0x0007 (PIN verification failed/timeout) where those codes remain canonical.

Table 2: Response to Challenge (osdp_CRAUTHR)

Size (bytes)	Name	Meaning	Value
1	CMND	Reply identifier.	0x82
2	TOTAL	Length of the complete response, least-significant byte first.	0x0000–0xFFFF
2	OFFSET	Offset of this fragment within the complete response, least-significant byte first.	0x0000–0xFFFF
2	DATA_LEN	Length of the fragment payload, least-significant byte first.	0x0000–0xFFFF
0–n	Response Data	Signature, encrypted block, or shared-secret material returned by the credential. The credential's BER-TLV structure is preserved exactly as received (status word excluded).	0x00–0xFF

2.1.3 Get PIV Data (osdp_PIVDATA)

OSDP 2.2 defined osdp_PIVDATA with a five-byte payload that selected a PIV object and optional element. The original text did not explicitly specify how to pad identifiers shorter than three bytes, how nested tags should be returned, or how to interpret the data offset. OSDP 2.3 clarifies these behaviors so that ACUs and PDs apply a single normative approach.

- The ACU **SHALL** transmit the PIV Object Identifier as three bytes encoded most-significant byte first. When the identifier contains fewer than three significant bytes, the ACU **SHALL** left-pad the value with 0x00 or 0x0000, as required, so that the transmitted field is exactly three bytes. The

PD **SHALL NOT** forward those pad bytes to the credential; it **SHALL** issue the ISO 7816-4 GET DATA APDU using only the significant identifier bytes. If VCI secure messaging is established, the PD **SHALL** use the appropriate secure channel automatically; no explicit selector is present in the command.

- The field previously described as “PIV element ID” is renamed PIV Tag. A tag value of 0x00 directs the PD to return the complete BER-TLV structure, including the outer 0x53 or 0x73 tag and length. Any non-zero value selects a single root-level tag within the object. The PD **MUST** remove the outer wrapper only when it is 0x53 or 0x73; if the outer wrapper has any other value, the PD **MUST** reject the command unless the tag value is 0x00. Tags **SHALL** be a single byte in the range 0x01–0xFF; multi-byte DER tags are not supported and **MUST** be retrieved by issuing a tag value of 0x00 and parsing the returned object at the ACU.
- The data offset now supports one- or two-byte encodings. PDs **SHALL** accept either form and determine which was used based on the command length. The offset applies to the selected payload (entire object when the tag is zero, or the chosen tag otherwise). PDs **SHALL** return `osdp_PDERROR` with error code 0x0123 when security prerequisites are not satisfied and 0x0124 when the requested object or tag is absent; see Subsection 2.2. If the credential is not present or is removed during processing, PDs may return error code 0x0127.

Table 3: Get PIV Data (`osdp_PIVDATA`)

Size (bytes)	Name	Meaning	Value
1	CMND	Command identifier.	0xA3
3	PIV Object Identifier	Three-byte identifier of the PIV data object per SP 800-73-4, encoded MSB first. Leading zero bytes SHALL be supplied by the ACU when the identifier is shorter than three bytes; the PD SHALL drop those pad bytes before issuing the APDU.	0x00–0xFF ¹
1	PIV Tag	Single-byte ASN.1 DER tag of a root-level element. Tag value 0x00 selects the entire object; non-zero tags select a root-level element and require the PD to strip the outer 0x53/0x73 wrapper, verify the length, and return only the tagged value. Multi-byte tags are not directly addressable and SHALL be retrieved by requesting tag 0x00.	0x00; or 0x01–0xFF
1–2	Data Offset	Byte offset within the requested content. Single-byte and two-byte encodings SHALL be accepted; the PD determines the format from the packet length.	0x00–0xFFFF

¹ Each position MAY carry any value from 0x00 to 0xFF; the ACU pads on the left with 0x00 when the identifier has fewer than three significant bytes.

2.1.4 PIV Data Reply (osdp_PIVDATAR)

The reply structure is unchanged from a field-count perspective but 2.3 aligns the semantics with the command refinements above:

- TOTAL, OFFSET, and DATA_LEN continue to be encoded LSB first; they now explicitly report the size, fragment base, and fragment length for the content selected by the osdp_PIVDATA request (entire object or chosen tag).
- CARD_DATA **SHALL** convey the exact bytes obtained from the credential after applying any tag stripping specified by the command. No additional framing or padding is permitted.

Table 4: PIV Data Reply (osdp_PIVDATAR)

Size (bytes)	Name	Meaning	Value
1	CMND	Reply identifier.	0x80
2	Total Length	Length in bytes of the complete data object or tagged element, least-significant byte first.	0x0000–0xFFFF
2	Fragment Offset	Offset in bytes of this fragment within the complete payload, least-significant byte first.	0x0000–0xFFFF
2	Fragment Length	Length in bytes of the data carried in this fragment, least-significant byte first.	0x0000–0xFFFF
0–n	Card Data	Requested data bytes exactly as returned by the credential. When a tagged element was selected, this sequence contains only the element's BER-TLV contents, excluding the selected tag and length.	0x00–0xFF

2.2 PD Error Codes for PIV

PD error codes are carried in `osdp_PDERROR` using two-byte codes (least-significant byte first). The common PD codes remain available (for example 0x0001 `SC_ERROR` with optional `SW1|SW2`, 0x0002 `SC_TIMEOUT`, 0x0004 `TLV_MISSING`, 0x0005 `INVALID_PARAM`, 0x0006 `SECURE_CHANNEL_REQUIRED`, 0x0007 `MP_INVALID`, 0x0008 `MP_OUT_OF_SEQUENCE`, 0x0009 `MP_OUT_OF_BOUNDS`). The following PIV-specific codes occupy the 0x0100–0x01FF range:

Table 5: Enhanced PIV Error Codes

Code	Meaning
0x0120	Function not supported by credential (or PD does not implement the command).
0x0121	Insufficient PD memory or storage to complete the requested operation.
0x0122	Insufficient credential memory or storage space.
0x0123	Credential security status not satisfied.
0x0124	Data object or tag not found.
0x0125	VCI not established.
0x0126	PIV Auto not configured.
0x0127	Credential not present or removed.
0x0128	Biometric type not supported.

2.2.1 PD Application Error Reply (osdp_PDERROR)

PDs **SHALL** use osdp_PDERROR to report application-level errors. The error code is two bytes, encoded least-significant byte first. When a card/APDU failure occurs (for example SC_ERROR or a PIV APDU failure), the PD **SHALL** append the ISO/IEC 7816 status word (SW1|SW2). Multiple errors **MAY** be reported by setting the error count and including additional code/SW pairs. Transport-level faults (framing, CRC, sequence) continue to use osdp_NAK.

Table 6: PD Application Error Reply (osdp_PDERROR)

Size (bytes)	Name	Meaning	Value
1	CMND	Command identifier.	0x84
2	Error Code	Two-byte error code, least-significant byte first.	0x0001–0x01FF
0 or 2	Status Word	Optional ISO/IEC 7816 status word for card/APDU failures; present only when defined by the command profile (for example PIV APDU errors).	–

2.3 Updated Function Codes

2.3.1 Function Code 12 – Smart Card Support

Function Code 12 advertises smart-card transport capabilities and profile support at the PD. The compliance byte uses the following bit assignments:

Table 7: Function Code 12 Compliance Flags

Mask	Meaning
0x01	PD supports transparent reader mode.
0x02	PD supports extended packet mode.
0x04	PD supports the OSDP 2.3 Enhanced PIV profile, including osdp_PIVDATA refinements and related commands.
0x08	PD supports OSDP 2.3 autonomous PIV operation (PIV Auto).

Bits not listed are reserved and **SHALL** be transmitted as zero.

2.3.2 Function Code 13 – Reader Interfaces

OSDP 2.2 defined Function Code 13 as a reader count indicator with a compliance byte fixed at 0x00. Enhanced PIV-capable PDs **SHALL** continue to populate the reader count field, but they **MAY** set bits within the compliance byte to advertise the downstream credential technologies they expose. A value of 0x00 remains valid and retains the legacy meaning of “interface presence unspecified.”

When a PD reports specific interfaces, it **SHALL** encode the compliance byte using the flags in Table 8. The reader count byte continues to report the number of attached credential interfaces.

Table 8: Function Code 13 Compliance Flags

Mask	Meaning
0x01	PD presents a contact card interface (ISO 7816 or equivalent).
0x02	PD presents a high-frequency contactless credential interface (13.56 MHz NFC).
0x04	PD presents a low-frequency credential interface (approximately 125 kHz prox).
0x08	PD supports Virtual Contact Interface (VCI) secure messaging for applicable modes.
0x10	PD presents a Bluetooth credential interface.
0x20	PD presents a barcode credential interface.
0x40	PD presents an ultra-high-frequency credential interface (UHF/RAIN RFID).

All other bits are reserved for future use and **SHALL** be transmitted as zero.

2.3.3 Function Code 14 – Biometric Support

Function Code 14 remains backward compatible with OSDP 2.2. A compliance value of 0x00, 0x01, 0x02, or 0x03 retains its original meaning and indicates that the PD does not support the OSDP 2.3

biometric command set. Devices that support the enhanced commands **SHALL** set Bit 7 (0x80). When Bit 7 is asserted, the remaining bits of the compliance byte are interpreted as a capability bitfield. Bits not explicitly defined in Table 9 are reserved and **SHALL** be transmitted as zero.

Table 9: Function Code 14 Compliance Flags

Mask	Meaning
0x80	PD supports the OSDP 2.3 biometric command set.
0x01	PD performs off-card fingerprint comparison using FIPS 201 BIO ANSI INCITS 378 templates.
0x02	PD performs on-card fingerprint comparison using FIPS 201 OCC processes.
0x04	PD performs off-card iris comparison aligned with FIPS 201 BIO guidance.
0x08	PD performs off-card facial comparison using ANSI INCITS 385-2004 images stored on the credential.

3 New or Enhanced PIV Commands

This section introduces commands defined exclusively for the OSDP 2.3 Enhanced PIV profile.

3.1 PIV Put Data (osdp_PIVPUTDATA)

The `osdp_PIVPUTDATA` command instructs the PD to deliver an ISO 7816-4 PUT DATA operation to the credential. The ACU **SHALL** construct the payload as a complete BER-TLV object, including the desired outer tag (0x7E for the Discovery Object, 0x7F61 for a BIT Group Template, 0x5C/0x53 sequences for other data objects, and so on). The PD **SHALL** forward the payload to the credential using CLA = 0x00, INS = 0xDB, P1 = 0x3F, and P2 = 0xFF. If VCI secure messaging is established, the PD **SHALL** select the secure channel automatically and adjust the CLA accordingly. Chained APDUs **SHALL** be generated by the PD when the payload exceeds a single-card APDU.

Table 10: PIV Put Data (osdp_PIVPUTDATA)

Size (bytes)	Name	Meaning	Value
1	CMND	Command identifier.	TBA
2	MpSizeTotal	Total size of the complete PUT DATA payload, least-significant byte first.	0x0000–0xFFFF
2	MpOffset	Offset of this fragment within the complete payload, least-significant byte first.	0x0000–0xFFFF
2	MpFragmentSize	Number of payload bytes carried in this fragment, least-significant byte first.	0x0000–0xFFFF
0–n	Data	BER-TLV payload to be written to the credential. This field already includes the outer object tag selected by the ACU.	0x00–0xFF

Responses A successful write is acknowledged with `osdp_ACK`. Failures **SHALL** use `osdp_PDERROR` with one of the PIV error codes defined in Subsection 2.2: 0x0120 when the function is not supported, 0x0121 when the PD lacks buffer space to stage the payload, 0x0122 when the credential has insufficient storage, 0x0123 when security conditions are not met, or 0x0127 when the credential is not present or is removed during the operation.

3.2 Card Status (osdp_CARDSTATUS)

The `osdp_CARDSTATUS` command requests the current credential context from the PD. The command code is to be assigned; PDs **SHALL** advertise support via Function Code 12 bit 0x04 (Enhanced PIV) before

controllers attempt to use this command. The payload selects the status record that should be returned (currently only the standard profile).

Table 11: Card Status Request (osdp_CARDSTATUS)

Size (bytes)	Name	Meaning	Value
1	CMND	Command identifier.	TBA
1	Status Type	Requested status record. 0x01 selects the standard PIV card status profile. Values 0x02–0x7F are reserved and shall elicit osdp_NAK. Values 0x80–0xFF are reserved for private use.	0x00–0xFF

PDs **SHALL** return osdp_PDERROR with an appropriate error code (for example 0x0003 unknown command, 0x0005 incorrect parameter) if the requested status type is not supported.

3.3 Card Status Reply (osdp_CARDSTATUSR)

In response to osdp_CARDSTATUS, the PD reports a ready-to-proceed credential state after completing its card-present workflow (power, application selection, and required reads). PDs **SHALL NOT** send osdp_CARDSTATUSR unsolicited on osdp_POLL. Controllers **SHOULD** continue to use legacy osdp_RAW poll replies as the single aggregated “card presented and data acquired” indication; PIN usage policy and VCI connectivity **SHALL NOT** be included in unsolicited poll indications. Detailed readiness (credential type, selected AID, cached data, and readiness flags) **SHALL** be returned only in solicited osdp_CARDSTATUSR.

Table 12: Card Status Reply (osdp_CARDSTATUSR)

Size (bytes)	Name	Meaning	Value
1	CMND	Reply identifier.	TBA
1	Detected Credential Type	Current card interface: 0x00 none, 0x01 ISO 7816 contact, 0x02 ISO 14443 contactless, 0x03 other ISO 7816 credential (for example DESFire, Seos), 0x04 low-frequency credential, 0x05 unknown type, 0x06–0x7F reserved, 0x80–0xFF private use.	0x00–0xFF
1	VCI Status	0x00 VCI not established, 0x01 VCI established. Values 0x02–0x7F reserved, 0x80–0xFF private use.	0x00–0xFF
1	VCI Trust Anchor ID	Identifier of the trust anchor used to establish VCI. 0x00 indicates no anchor in use.	0x00–0xFF
2	PIN Usage Policy	PIN Usage Policy bytes (tag 0x5F2F) returned verbatim when available. A value of 0xFFFF indicates the policy is not present or has not yet been read.	0x0000–0xFFFF
1	Selected AID Length	Length (0–16) of the selected AID. 0x00 indicates no application is currently selected.	0x00–0x10
0–16	Selected AID	Application Identifier read from the credential (for example from the FCI template). Present only when the length is non-zero.	0x00–0xFF

A PD that cannot provide the requested status **SHALL** return `osdp_PDERROR` and omit the reply. Successful replies include the entire selected AID (when available) and any PIN Usage Policy bytes read from the credential's Discovery Object (tag 0x5F2F) only after the PD has completed its preparation. When no PIV PIN Usage Policy is present, the PD **SHALL** report a length of zero and omit the bytes. The VCI fields describe stable readiness (for example whether a secure channel is established) and whether the presented credential requires a pairing code; transient negotiation state **SHALL NOT** be reported.

3.4 PIV Secure PIN Entry (osdp_PIVSPE)

The osdp_PIVSPE command instructs the PD to perform an ISO/IEC 7816 VERIFY operation on the credential, allowing the cardholder to enter a PIN directly on the reader. The ACU is responsible for conveying a clear user indication that secure PIN entry is in progress (for example, via displays or annunciators). While osdp_PIVSPE is active, the PD **SHALL** suppress all osdp_KEYPAD reports to the ACU and terminate SPE mode only after returning a result via osdp_PIVSPER. The command code is to be assigned.

Table 13: PIV Secure PIN Entry (osdp_PIVSPE)

Size (bytes)	Name	Meaning	Value
1	CMND	Command identifier.	TBA
1	SPE Mode	Secure PIN Entry operating mode. See Table 14.	0x00–0xFF
1	Start Timeout	Seconds allowed to begin key entry (0 = no timeout).	0x00–0xFF
1	Inter-key Timeout	Seconds allowed between keystrokes (0 = no timeout).	0x00–0xFF
1	Flags	Behaviour modifiers. See Table 15.	0x00–0xFF
2	Cache Validity	PIN cache validity in seconds (LSB first). A value of 0x0000 disables caching.	0x0000–0xFFFF

Table 14: SPE Mode Values

Value	Description
0x00	Status only – report current SPE state without initiating entry.
0x01	Normal – single PIN verification with the credential present (contact).
0x02	VCI pre-cache – collect a PIN prior to credential presentation for VCI use.
0x03	VCI execute – perform PIN verification over VCI using a cached PIN.
0x04	PIN caching – cache PIN for subsequent operations (non-FICAM use cases).
0x05	Clear/abort – clear any cached PIN and abort the current operation.
0x06–0x7F	Reserved for future use (set to zero).
0x80–0xFF	Private use.

Table 15: SPE Flag Values

Mask	Meaning
0x01	Allow retry continuation when the credential remains present after a PIN-entry failure.
0x02	Require an explicit cache-clear command; do not auto-clear cached PINs on credential removal.
0x04	Skip pairing-code verification when the credential and policy permit VCI without a pairing code.
0x08	Auto-execute VCI verification when the credential is presented after a successful pre-cache (Mode 0x02).
0x10–0x80	Reserved and SHALL be transmitted as zero.

Controllers select an SPE Mode to match the desired workflow (status query, contact PIN verification, VCI pre-cache/execute, PIN caching, or abort). Modes that rely on VCI (for example 0x02 and 0x03) require the PD to have established VCI beforehand; otherwise the PD **SHALL** return `osdp_PDERROR 0x0125`. Start Timeout and Inter-key Timeout values of zero disable the respective timer. Cache Validity is expressed in seconds (LSB first); a value of zero disables PIN caching. If a mode requests PIN caching, the PD **SHALL NOT** transmit the PIN to the ACU.

A successful request is acknowledged with `osdp_ACK`. The PD **SHALL** return `osdp_PDERROR` with one of the following codes when a failure occurs: 0x0120 (function not supported), 0x0005 (incorrect parameter), 0x0123 (security status not satisfied), 0x0125 (VCI not established for VCI-specific modes), or 0x0127 (credential not present or removed). After issuing `osdp_PIVSPE`, the ACU **SHALL NOT** send other PIV commands until a corresponding `osdp_PIVSPER` reply is received.

3.5 PIV Secure PIN Entry Reply (`osdp_PIVSPER`)

The `osdp_PIVSPER` reply reports the outcome of the most recent `osdp_PIVSPE` command. The PD sends `osdp_PIVSPER` as a response to `osdp_POLL` once SPE completes. The reply command code is to be assigned.

Table 16: PIV Secure PIN Entry Reply (`osdp_PIVSPER`)

Size (bytes)	Name	Meaning	Value
1	CMND	Reply identifier.	TBA
1	Result	Outcome of the secure PIN entry operation. See Table 17.	0x00–0xFF
1	Attempts Remaining	Match attempts remaining when the result is PIN mismatch; otherwise transmitted as 0x00.	0x00–0xFF

Result codes follow Table 17. When the result indicates a PIN mismatch, the Attempts Remaining byte

conveys the number of retries still permitted; otherwise the field is reserved and **SHALL** be transmitted as zero.

Table 17: PIVSPE Result Codes

Value	Description
0x00	Success – PIN verified.
0x01	Timeout – PIN entry did not begin within the allotted time.
0x02	Timeout – inter-digit interval exceeded.
0x03	PIN mismatch.
0x04	Credential removed during entry.
0x05	Credential PIN blocked.
0x06	Wrong interface for the requested mode.
0x07	Incorrect parameters.
0x08	Security condition not met.
0x09–0xFF	Reserved for future use.

3.6 PIV Biometric Match (osdp_PIVBIOMATCH)

The osdp_PIVBIOMATCH command instructs the PD to acquire a live sample and compare it to the specified biometric template from the active application. Depending on the Match Mode, the comparison occurs either on the credential (on-card match) or within the PD (off-card match). The command code is to be assigned.

Table 18: PIV Biometric Match (osdp_PIVBIOMATCH)

Size (bytes)	Name	Meaning	Value
1	CMND	Command identifier.	TBA
1	Biometric Type	Selects the template type to compare. See Table 19.	0x00–0xFF
1	Match Mode	0x00 off-card match; 0x01 on-card match.	0x00–0x01

Table 19: Biometric Type Values

Value	Description
0x01	Cardholder fingerprint template.
0x02	Cardholder facial image.
0x04	Cardholder iris template.

Controllers set Match Mode to select off-card or on-card comparison. Off-card flows require the PD to retrieve the appropriate template and, where applicable (e.g., TWIC contactless), to have a valid privacy key loaded via `osdp_PIV_LOADTPK`. On-card flows require the credential to support OCC. If prerequisites are not met, the PD **SHALL** return `osdp_PDERERROR` with one of the PIV error codes (for example, 0x0120 function not supported by credential, 0x0123 credential security status not satisfied, 0x0125 VCI not established, 0x0127 credential not present/removed, or 0x0128 biometric type not supported).

3.7 PIV Biometric Match Reply (`osdp_PIVBIOMATCHR`)

After `osdp_PIVBIOMATCH` completes, the PD reports the result as a poll response. The reply command code is to be assigned.

Table 20: PIV Biometric Match Reply (`osdp_PIVBIOMATCHR`)

Size (bytes)	Name	Meaning	Value
1	CMND	Reply identifier.	TBA
1	Result	Outcome of the biometric match. See Table 21.	0x00–0xFF
32	Hash (optional)	Present on success when required: 32-byte SHA-256 hash of the biometric tag 0xBC data field. Omitted on failure results.	0x00–0xFF

Table 21: PIVBIOMATCH Result Codes

Value	Description
0x00	Biometric match success (off-card match).
0x01	Biometric match timeout (no entry started).
0x02	Biometric match failed (no match).
0x03	Credential removed or not present.
0x04	PIN verification required.
0x05	PIN verification failed.
0x06	PIN verification timeout.
0x07	CHUID mismatch.
0x08	PD error during biometric acquisition.
0x09	On-card match success.
0x0A–0x7F	Reserved for future use.
0x80–0xFF	Private use.

3.8 TWIC Privacy Key Load (osdp_PIV_LOADTPK)

The `osdp_PIV_LOADTPK` command instructs the PD to cache a TWIC Privacy Key (TPK) for use with subsequent biometric operations against a TWIC credential. The command code is to be assigned. PDs **SHALL** advertise support for TWIC biometric processing via Function Code 14 before controllers attempt to use this command.

Table 22: TWIC Privacy Key Load (osdp_PIV_LOADTPK)

Size (bytes)	Name	Meaning	Value
1	CMND	Command identifier.	TBA
1	Cache Timeout	Number of seconds to retain the cached TPK. 0x00 clears any cached key immediately; 0x01–0xFF permit caching for up to 255 seconds.	0x00–0xFF
1	Control Flags	Bit 0 set requests automatic clearing when the credential is removed; all other bits SHALL be zero.	0x00–0xFF
32	TPK	Trusted Processing Key bytes to cache for the next TWIC biometric operation.	0x00–0xFF

Controllers set Cache Timeout to define how long (in seconds) the PD may retain the key. A value of 0x00 clears any cached TPK immediately. Values 0x01–0xFF allow caching for up to 255 seconds. Bit 0 of

Control Flags directs the PD to clear the cached TPK when the credential is removed; all other bits are reserved and **SHALL** be transmitted as zero. If the PD does not support TWIC TPK caching it **SHALL** return `osdp_PDERROR 0x0003`. Invalid payloads (for example reserved bits set) **SHALL** elicit `osdp_PDERROR 0x0005`.

3.9 PIV VCI Pairing Code Transmit (`osdp_PIV_XMITPAIRING`)

The `osdp_PIV_XMITPAIRING` command delivers an eight-digit pairing code to the PD for use during Virtual Contact Interface (VCI) establishment. The command code is to be assigned. Controllers **SHALL** **NOT** issue this command unless a PIV application is currently selected and VCI mode is active.

Table 23: PIV VCI Pairing Code Transmit (`osdp_PIV_XMITPAIRING`)

Size (bytes)	Name	Meaning	Value
1	CMND	Command identifier.	TBA
8	Pairing Code	Eight ASCII digits ('0'–'9') representing the VCI pairing code associated with the currently selected PIV application.	0x30–0x39

The pairing code is encoded as eight ASCII digits ('0'–'9'). A PD that has not yet established VCI **SHALL** return `osdp_PDERROR 0x0125` (VCI not established). If no PIV application is selected, the PD **SHALL** return `osdp_PDERROR 0x0005`. PDs **SHALL** clear the stored pairing code when VCI is torn down or when the active PIV mode changes.

3.10 Set PIV Mode (`osdp_PIVMODE`)

The `osdp_PIVMODE` command defines the operating context for PIV, TWIC, and related credential transactions. OSDP 2.3 replaces the legacy “Application + AID” fields with a payload that begins with a required Global Interface Mask byte followed by a TLV list of configuration descriptors so controllers can specify multiple application profiles, preferred interfaces, and optional Application Identifiers. The command code remains to be assigned. PDs **SHALL** advertise support via Function Code 13 bit 0x04.

Table 24: Set PIV Mode (osdp_PIVMODE)

Size (bytes)	Name	Meaning	Value
1	CMND	Command identifier.	TBA
1	PIV Mode Flags	Global behaviour flags (Section 3.10).	0x00–0xFF
2	MpSizeTotal	Total size of the complete configuration payload, least-significant byte first.	0x0000–0xFFFF
2	MpOffset	Offset of this fragment within the configuration payload, least-significant byte first.	0x0000–0xFFFF
2	MpFragmentSize	Length of the fragment payload, least-significant byte first.	0x0000–0xFFFF
1	Config Count	Number of configuration entries (1–32).	0x01–0x20
1	Global Interface Mask	Bitmask limiting interfaces available to all configurations (bit 0 contact, bit 1 contactless, bit 3 Bluetooth; bit 2 reserved and set to zero).	0x00–0x0B
0–n	TLV Data	Sequence of Configuration Entry TLVs (Tag 0x01) that follow the mask.	–

The PIV Mode Flags byte enables global behaviours such as autonomous operation and VCI usage (Table 25). Bits not listed remain reserved and **SHALL** be transmitted as zero.

Table 25: PIV Mode Flags

Mask	Meaning
0x01	Enable autonomous PIV (PIV Auto). The PD interrogates credentials, establishes VCI when possible, and advances through the authentication flow without explicit ACU prompts.
0x02	Permit VCI usage when available.
0x04	Enforce FICAM strict mode (PIV applications only).

After the flags, the payload uses standard multi-part framing fields and a configuration count (1–32). The

count is followed immediately by the required Global Interface Mask byte. Each configuration appears as a TLV:

- **Global Interface Mask (required)** — One-byte mask (no tag) that limits the interfaces the PD may use for all configurations. Bit assignments follow Function Code 13 (bit 0 contact, bit 1 contactless, bit 3 Bluetooth); bit 2 is reserved and **SHALL** be zero. A value of 0x00 indicates no global restriction.
- **Configuration Entry (Tag 0x01)** — Contains the data-model identifier, a per-entry interface mask, the AID length, and the optional AID:
 - Data Model (0x01 PIV, 0x02 TWIC, values 0x04–0x7F reserved, 0x80–0xFF private use).
 - Interface Mask — Uses the same bit positions as the global mask; 0x00 means “any interface permitted by the global mask.”
 - AID Length (0–16) and AID bytes when provided. The legacy AID list referenced in earlier proposals is informational; it is not part of the official OSDP 2.2 specification.

PDs process configuration entries in order, applying the global mask (when present) before honouring each entry’s mask. Invalid masks, unknown data-model identifiers, or malformed TLVs **SHALL** result in `osdp_PDERROR 0x0005`. Controllers **SHALL** send the command over a secure channel where required; the PD **SHALL** return `osdp_PDERROR 0x0006` if the necessary security conditions are not satisfied. Multi-part framing errors result in `osdp_PDERROR 0x0007`. If the PD cannot cache the requested configuration set, it **SHALL** respond with `osdp_PDERROR 0x0121`.

When PIV Auto is enabled, PDs interrogate the credential on presentation, establish VCI automatically when possible, and advance through authentication without explicit ACU prompts. If a pairing code is required, the PD signals the state change via a solicited `osdp_CARDSTATUSR` (not via unsolicited poll); the ACU then provides the code using `osdp_PIV_XMITPAIRING`. Commands that rely on PIV Auto **SHALL** return `osdp_PDERROR 0x0126` when the feature has not been configured.

3.11 VCI Trust Anchor Load (`osdp_PIV_VCILOADTA`)

The `osdp_PIV_VCILOADTA` command loads, replaces, or deletes a Virtual Contact Interface (VCI) trust anchor. Trust anchors must be cached before a PD can verify card certificates during VCI establishment. The command code is to be assigned.

Table 26: VCI Trust Anchor Load (osdp_PIV_VCILOADTA)

Size (bytes)	Name	Meaning	Value
1	CMND	Command identifier.	TBA
1	Operation	0x00 load/replace anchor; 0x01 delete anchor; 0x02 query anchor inventory.	0x00–0x02
1	Trust Anchor ID	Controller-assigned identifier for the anchor (1–128).	0x00–0xFF
8	Issuer Identifier Number	Leftmost eight bytes of the subjectKeyIdentifier from the issuing content-signing certificate.	0x00–0xFF
2	Anchor Length	Total size of the anchor payload, least-significant byte first. Set to 0x0000 when deleting or querying inventory.	0x0000–0xFFFF
2	MpSizeTotal	Total size of the complete anchor data, least-significant byte first. Set to 0x0000 when deleting or querying inventory.	0x0000–0xFFFF
2	MpOffset	Offset of this fragment within the anchor data, least-significant byte first. Set to 0x0000 when deleting or querying inventory.	0x0000–0xFFFF
2	MpFragmentSize	Length of the data block in this fragment, least-significant byte first. Set to 0x0000 when deleting or querying inventory.	0x0000–0xFFFF
0–n	Anchor Data	Anchor bytes (DER or equivalent) present when loading. Omitted for delete and query operations. Subsequent fragments (load only) contain only this field.	0x00–0xFF

The metadata block (operation, identifier, IIN, anchor length) appears only in the first fragment where `MpOffset = 0`; subsequent fragments carry additional anchor bytes. Controllers set `Operation` to 0x00 to load or replace the anchor identified by `Trust Anchor ID`. Set `Operation` to 0x01 with `Anchor Length` 0x0000 to delete the anchor. The `Issuer Identifier Number` (IIN) is the leftmost eight bytes of the subject-

tKeyIdentifier in the content signing certificate used to protect the credential. The Anchor Length declares the total size of the anchor payload; controllers shall send the exact number of bytes in one or more fragments.

PDs **SHALL** reconstruct the anchor from the fragments before storing it. A successful operation returns `osdp_ACK`. Failures return `osdp_PDERROR` with:

- 0x03 when the PD does not support trust-anchor caching.
- 0x05 for malformed metadata (unknown operation, reserved bits, invalid lengths).
- 0x07 when multi-part sequencing fails.
- 0x21 when the PD lacks storage space for the anchor.

PDs are not required to maintain a real-time clock; they need only verify that credential certificates presented later fall within the trust anchor's validity period. Implementers may perform stricter date checks if desired. PDs may retain either the full anchor blob or a compact record containing the public key, issuer identifier, validity period, and anchor number so long as the supplied IIN remains available for matching.

Trust anchors may be stored in volatile or non-volatile memory at the PD's discretion. Controllers should not assume retention beyond the device's documented behaviour.

3.12 VCI Trust Anchor Status (`osdp_PIV_VCITASTATUS`)

The `osdp_PIV_VCITASTATUS` command enumerates cached trust anchors and reports remaining storage capacity. The command code is to be assigned. The request carries no payload. The reply uses the standard multi-part fields (`TOTAL`, `OFFSET`, `DATA_LEN`) so PDs can stream the inventory across one or more fragments.

Table 27: VCI Trust Anchor Status Reply (`osdp_PIV_VCITASTATUSR`)

Size (bytes)	Name	Meaning	Value
1	CMND	Reply identifier.	TBA
2	TOTAL	Length of the complete status payload, least-significant byte first.	0x0000–0xFFFF
2	OFFSET	Offset of this fragment within the status payload, least-significant byte first.	0x0000–0xFFFF
2	DATA_LEN	Length of the data block in this fragment, least-significant byte first.	0x0000–0xFFFF
1	Total Anchor Count	Number of cached trust anchors.	0x00–0xFF
2	Available Bytes	Remaining storage capacity for trust anchors, least-significant byte first.	0x0000–0xFFFF
0–n	Anchor Records	Zero or more records. Each record contains: Trust Anchor ID (1 byte), Issuer Identifier Number (8 bytes), Anchor Length (2 bytes LSB first), Attributes (1 byte, reserved). Records that span fragments continue in subsequent replies.	–

The PD first reports the total number of anchors and the remaining free bytes. Each record thereafter lists a trust anchor by ID, IIN, anchor length, and reserved attributes (transmitted as zero). Error responses follow the standard pattern: `osdp_PDError 0x0003` for unsupported command, `osdp_PDError 0x0007` for multi-part errors.

Table 28: Trust Anchor Record Structure

Field	Size	Meaning	Value Range
Trust Anchor ID	1 byte	Identifier assigned by the ACU; reused to replace or delete an anchor.	0x00–0xFF
Issuer Identifier Number (IIN)	8 bytes	Leftmost eight bytes of the subjectKeyIdentifier from the issuing content-signing certificate.	0x00–0xFF
Anchor Length	2 bytes (LSB first)	Total size of the stored anchor or compact representation.	0x0000–0xFFFF
Attributes	1 byte	Reserved for future use; transmitted as zero.	0x00

A References

- Security Industry Association, Open Supervised Device Protocol (OSDP) v2.2, 2020.
- National Institute of Standards and Technology, NIST Special Publication 800-73-4, Interfaces for Personal Identity Verification (PIV) — Part 2: PIV Card Application Card Command Interface, February 2015.
- National Institute of Standards and Technology, Federal Information Processing Standards Publication 201-3, Personal Identity Verification (PIV) of Federal Employees and Contractors, January 2022.
- U.S. General Services Administration, Federal Identity, Credential, and Access Management (FICAM) Architecture, Revision 3, 2023.
- Open Physical, High Speed Autonomous PIV: Concept Paper, 2024.

B Sample osdp_PIVDATA Requests (Informative)

This appendix illustrates representative requests that conforming ACUs may issue using the clarified osdp_PIVDATA command. Object identifiers align with NIST SP 800-73-4 Part 2, Table 6. Each example shows the bytes that appear in the OSDP payload after the CMND field; the PD strips leading zeroes before constructing the ISO 7816-4 APDU.

Cardholder Unique Identifier (CHUID) Object identifier 0x5FC102, tag 0x00, offset 0x0000. The PD transmits GET DATA for tag 0x5FC102 and returns the complete CHUID BER-TLV so the ACU can validate the FASC-N, GUID, and expiration fields.

Card Capability Container (CCC) Object identifier 0x5FC107, tag 0x00, offset 0x0000. Use this request during device discovery to read feature flags (contact/contactless support, optional biometrics) prior to issuing PIV Mode updates.

Discovery Object Object identifier 0x00007E, tag 0x00, offset 0x0000. The PD forwards tag 0x7E with its nested data so the ACU can parse the application identifier (tag 0x4F) and PIN Usage Policy (tag 0x5F2F). To retrieve only the AID, repeat the command with tag 0x4F.

Biometric Information Templates (BIT) Group Template Object identifier 0x007F61, tag 0x00, offset 0x0000. Large responses for fingerprint or iris templates require multiple osdp_PIVDATAR fragments; the TOTAL/OFFSET/DATA_LEN fields support reconstruction.

Selective Tag Retrieval Object identifier 0x00007E, tag 0x4F, offset 0x0000. The PD returns only the PIV Application Identifier nested inside the Discovery Object. Tags that occupy more than one byte (for example 0x7F61) are outside the scope of this selector and must be obtained by requesting tag 0x00.

C Deprecated osdp_GENAUTH

The legacy osdp_GENAUTH command is deprecated and expected to be removed in a future revision of this specification. Using a PIV credential to decrypt or process arbitrary data is unsafe and unnecessary for credential verification; implementations **SHOULD** use osdp_CRAUTH flows instead. Controllers and PDs **SHOULD** plan migrations away from osdp_GENAUTH.